Potential of Impedance Spectroscopy for Real-Time Assessing of Food Quality

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Reliable information about food state and freshness throughout production processes and until consumption is very important for product quality, consumer safety and export of goods. Regulations worldwide protect consumers and encourage the production of safe, nutritious and affordable aliments. Several measurement methods for food quality assessment can be used in the laboratory e.g., chemical analysis, mechanical methods, optical methods, x-ray measurements and nuclear magnetic resonance. Even if they are precise, these methods remain useless for real-time application.

Impedance spectroscopy has shown a high potential for real-time use, like in electronic tongues, and not necessarily under laboratory conditions [1]-[3], because changes of the complex impedance of food give information about its composition and help to follow changes of its state. After an introduction about impedance spectroscopy, this paper introduces some examples of its application to food characterization, such as meat freshness, milk dilution and water contents in edible oils.

Impedance Spectroscopy

Impedance spectroscopy is a system identification method that relies on measurements of the complex impedance of materials and systems at different excitation frequencies which allow information regarding their electrical and electrochemical properties to be extracted. This can be performed by modeling [4] or mathematical methods, such as data mining, principal component analysis [5], neural networks [6] and fuzzy logic [7]. For modeling, it is necessary to understand the physical and chemical phenomena taking place and to formulate the necessary model equations. The extraction of information is mainly related to model parameters, which are directly related to the corresponding physical phenomena. For mathematical methods, a big database of experiments is necessary to build the necessary basis for signal processing and extraction of information. Suitable features extraction and classification methods should be developed to discriminate between the different classes.

Impedance measurement has decisive benefits compared to other characterization methods, as it is experimentally efficient and can be realized in low cost, embedded and portable systems. It delivers more information than only resistive, capacitive or inductive measurements, because it uses both real and imaginary parts of the complex impedance. It provides the possibility to separate effects that have linear independent dispersions and dominate in different frequency ranges. Impedance spectroscopy can be used for non-invasive measurements, as it uses small amplitudes of excitation signal. It is capable of achieving real-time measurement in a relatively short measurement time as it can use wide frequency band signals. Especially for food, impedance spectroscopy is often preferred to optical methods, particularly in cases where measurements should be carried out deep in the material or when other substances interfere with the measurement quantity at that frequency. Mainly for these reasons, impedance spectroscopy is gaining increasingly in importance in biological and medical fields and will play a big role in the food industry in the future.

Selected Applications

Meat Freshness Assessment

Different meat scandals throughout the last decades have shown the importance of meat quality assessment during production, transportation and storage. Specifically, exposure to inadequate temperature leads to meat deterioration. Corruption has often included the wrong labels being put on old meat or selling some meat types in other, more expensive meat categories. Impedance spectroscopy can play a big role against this kind of corruption by developing corresponding test methods, which are able to differentiate between different states of muscles in real time.

To investigate possibilities for the realization of such measurement devices, bio-impedance measurements have been carried out for different muscles of beef, including Longisimus Dorsi (LD), Rectus Abdominus (RA) and Semi Membranous (SM) tissues (Fig. 1) [1], [6]. Samples are

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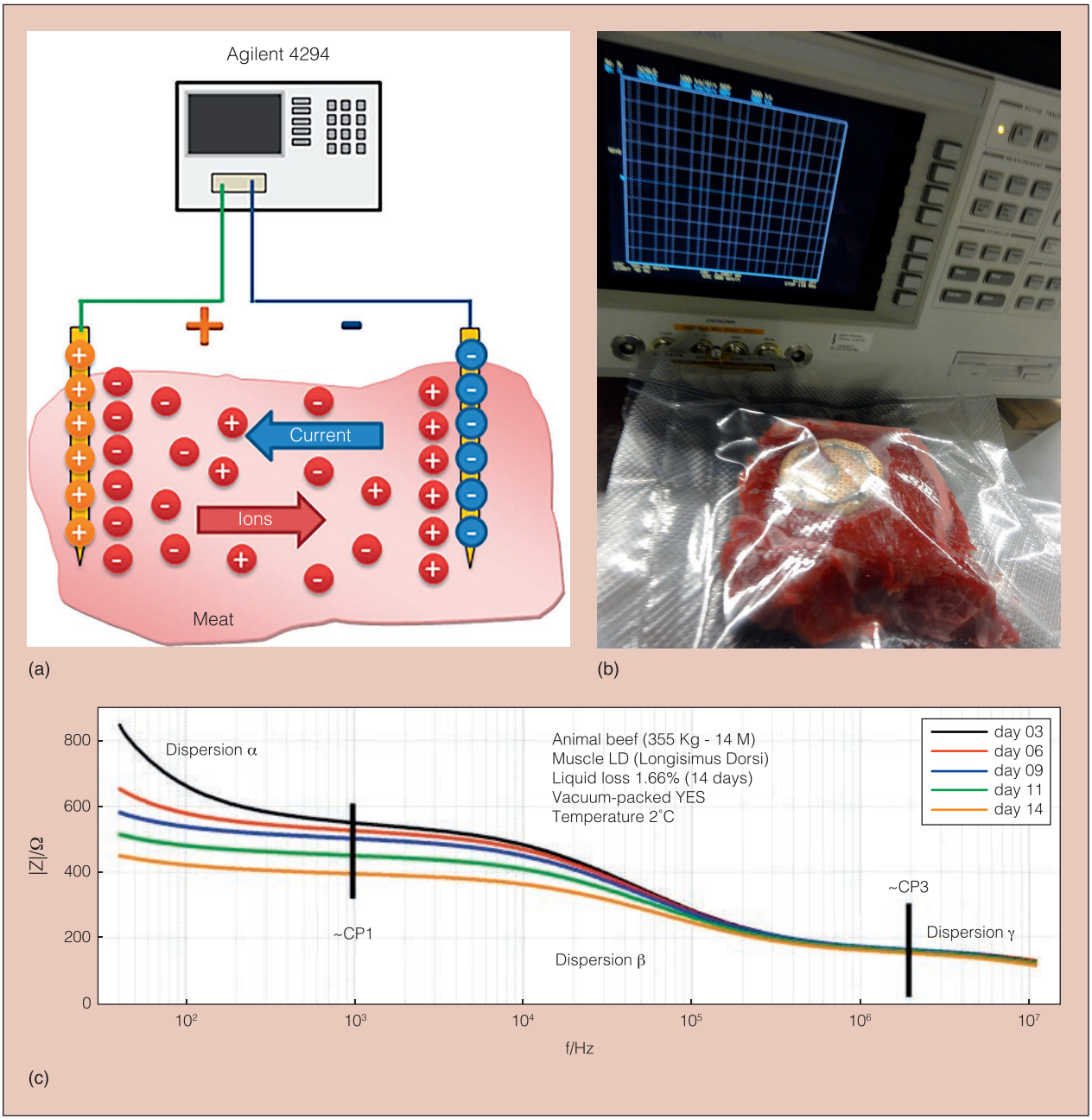

Fig. 1. Bio-impedance measurement. (a) Measurement setup. (b) Meat measurement in laboratory. (c) Impedance results.

vacuum-packed to avoid the appearance of bacteria on the surface and water losses during the experiments [8], and meat is stored at a temperature of 2 °C to maintain similar conditions used in the meat selling sector. Meat measurements are recorded on the laboratory impedance analyzer Agilent 4294 A, where the scanning 201 frequency follows a logarithmic law ranging from 40 Hz to 110 MHz and a sinusoidal signal is used as the excitation signal with a fixed voltage of 500 mV (Fig. 1a).

Many challenges need to be overcome during the measurement of muscle impedance to guarantee reproducible measurements (Fig. 1b). An electrode configuration is used to avoid the influence of anisotropy and to realize a good contact to the biological material in spite of its fibrous structure [8]. The measurement probe is inserted into the meat and consists of one central cylindrical electrode and eight surrounding needle electrodes made of gold plated steel (Ø 1 mm) to avoid corrosion.

The complex impedance of biological muscles shows in general three areas of dispersion, which are the α dispersion at low frequencies, β dispersion at middle frequencies and γ dispersion at higher frequencies [1]. The β dispersion corresponds to radio frequency and is strongly dependent on the dielectric properties of the cell membranes and their interactions with the extra and intracellular electrolytes. As the current passes through the cell membranes at these frequencies, it includes

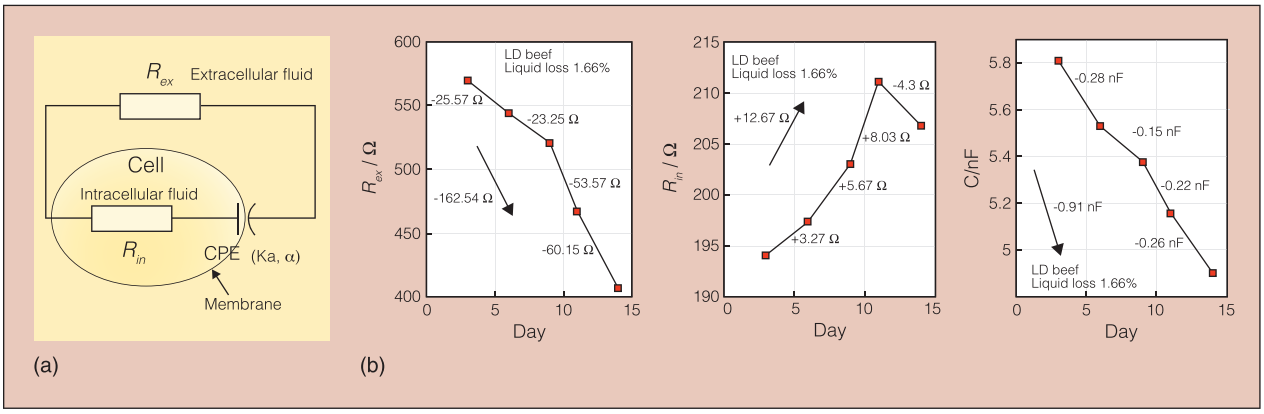

Fig. 2. (a) Fricke Cole Cole model. (b) Model parameters evolution over time for the muscle (LD Beef: Longisimus Dorsal).

information referring to the behavior of the cell membranes' integrity, which changes significantly with aging. One of the phenomena that contribute to the β dispersion is the Maxwell-Wagner effect due to the interfacial relaxation process occurring in systems where the electric current passes at the interface between two different materials.

Based on the theories of Fricke and using the Cole-Cole empirical equation, the model is composed of a resistive element R_{ex} that represents extracellular resistance placed in parallel with the resistive element R_{in} that represents intracellular resistance in series with the constant phase element (CPE) that corresponds to the insulating cell membranes (Fig. 2a).

$$Z_{\mathrm{model}} = \frac{1}{\dfrac{1}{R_{ex}} + \dfrac{1}{\dfrac{1}{\left(k_a\left(j.\omega\right)^{\alpha}\right)} + R_{in}}} \tag{1}$$

The model parameters of CPE, k_a and α are converted into a capacitance by (2) to get a physical explanation about the mechanisms that occurs during meat aging, where ω_m'' is the frequency corresponding to the local minimum of the imaginary part of the spectrum in the part in the range of the β dispersion.

$$C = k_a\left(\omega_m''\right)^{\alpha-1} \tag{2}$$

The model parameters show a good fitting to the expected postmortem behavior of the meat. The shrinkage of fibers during the early postmortem period causes the release of inner-cellular fluid to the extracellular fluid. This leads to an increase of the conductivity of the extracellular fluid and a decrease in the conductivity of the intracellular fluid. This behavior can be observed for the extracellular resistance and intracellular resistance (Fig. 2b). The measurement results for different muscles of beef or veal in different periods confirm the effectiveness of the bio-impedance measurement of meat [1]. These novel results show that testing devices for meat quality assessment can be realized based on impedance spectroscopy, to allow real time tests along the production process that benefit consumers.

Milk Water Concentration

Because of the lack of strict vigilance by food safety authorities, especially in developing countries, milk is often subject to fraud and adulteration. Water is the most common adulterant used, which decreases the nutritional value of milk and causes serious hygienic problems if the water is contaminated. Impedance spectroscopy provides the possibility to detect fraud in the milk sector. This method uses measurement of the impedance of a sensor dipped in milk and deduces its characteristics from the measured impedance.

To investigate the feasibility of using this method for milk state assessment, impedance measurements are obtained from a sensor consisting of two parallel platinum electrodes dipped completely in milk, so that parasitic factors can be avoided. Before each experiment, the sensor is cleaned with ethanol and rinsed with deionized water. Each experiment is repeated many times to check the measurement reproducibility. The results show that it was possible to detect even the presence of small portion of deionized water in milk [3]. Milk is an acidic solution with a pH value about 6.7 at room temperature, and the addition of water shifts the pH value towards that of a more neutral solution, so that the conductivity of the solution decreases. This explains the increase of the sensor impedance by an increase of water content (Fig. 3).

Impedance spectrum can be modelled as a resistance R_1 that represents a small serial resistance due to the measurement setup, a resistance R_2 that represents the electrolytic conductivity of the milk, a capacitance C that represents the capacitance of the sensor and a constant phase element (CPE) that represents diffusion effects. The resistance R_2 and the capacitance C increase linearly with the increase of the deionized water content because the increase of the deionized water content makes the mixture less conductive. The results show that addition of 5% of water causes a significant change in the measured impedance.

Water Content of Edible Oil

Impedance spectroscopy can be also considered for primarily dielectric non-conductive materials, such as oils, to detect their deterioration with a better precision. For example, frying is widely used in food preparation. It is a process where the food

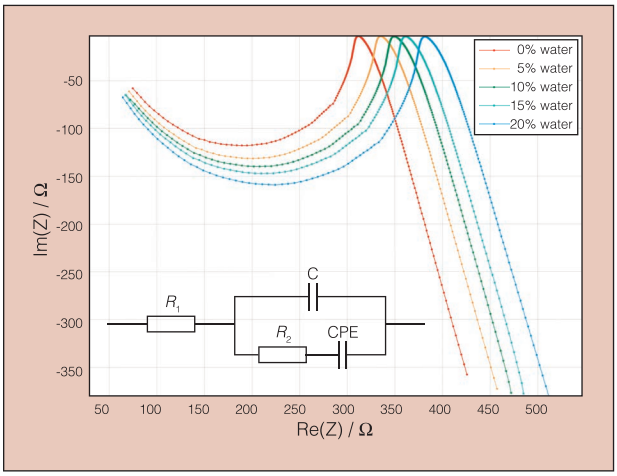

Fig. 3. Nyquist plots of the milk samples impedance and the corresponding equivalent circuit.

is submerged into hot oil at a high temperature of 150 °C to 180 °C, and this causes a degradation in the oil quality [10] due to the chemical processes such as thermal oxidation, hydrolysis and polymerization that produce a considerable number of harmful components [11].

Various techniques can be used to assess quality of oils like colorimetric reactions and chemical analysis. However, these techniques are complex, time consuming and must be carried out in a chemical laboratory by experts. Alternatively, the measurement of impedance spectroscopy can be simply done, realizing a continuous evaluation of oil quality under frying conditions.

To determine feasibility, four different oils were experimentally investigated: two sunflower and two olive oil samples. Each oil was heated to 180 °C using a laboratory hot plate. Four samples of each oil type were investigated: fresh, 10 hours heated, 20 hours heated and 30 hours heated. The change of the oil quality is characterized by measuring different properties, such as the change in acidity, total polar compounds and impedance at different heating times. The measurement of the acidity was done according to the standard ISO 660 based on the titration of oil with a base solution (KOH). The measurement of the total polar components was done according to the standard ISO 8420 based on the column chromatography technique. This technique is used to separate analytic mixtures of two or more compounds based on differences in polarity.

The measurement setup for impedance spectroscopy consists of a precise impedance analyzer connected to a capacitive sensor that consists of 45 parallel plates and has 136-pF capacitance in air. It provides a resolution of 1.36 pF of capacitive change for 0.01 change of dielectric constant. The impedance of the sensor submerged in oil is measured over the frequency range of 40 Hz to 2 MHz by applying a sinusoidal excitation voltage of 500 mV. The results are shown in Fig. 4a.

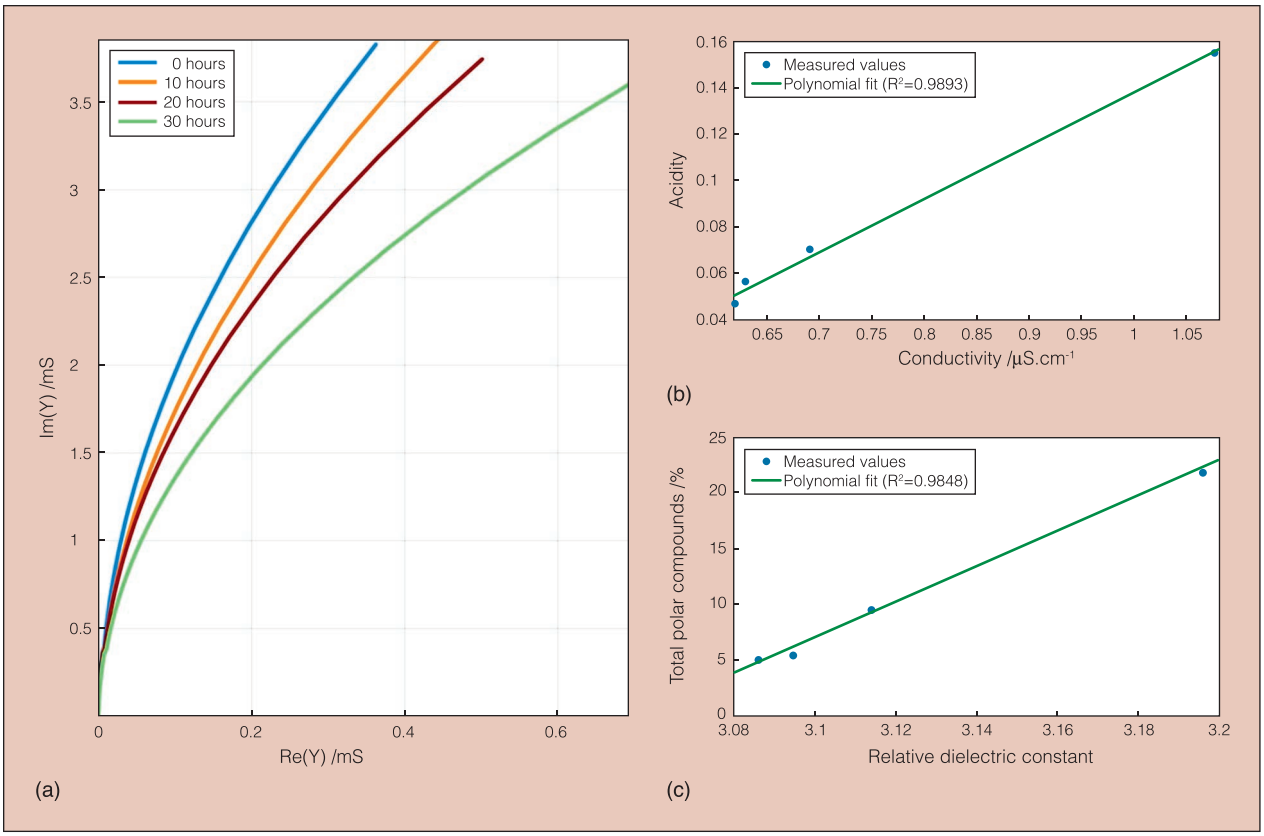

Fig. 4 (a) Nyquist admittance diagram for sunflower oil at different heating time. (b) Correlation between acidity and conductivity for sunflower oil. (c) Correlation between total polar components and dielectric constant for sunflower oil.

For physical modeling, edible oil can be described by an equivalent circuit of a lossy capacitance formed by a capacitor C in parallel to a conductance G.

$$\underline{Y} = G + jC\omega = Re\left(\underline{Y}\right) + j.Im\left(\underline{Y}\right) \tag{3}$$

In (3), C and G take into consideration the dielectric constant ε' and the conductivity σ of the sample, respectively. The dielectric constant can be calculated from the imaginary part of the admittance (4), and the conductivity can be calculated from the real and imaginary parts of the admittance (5) where C_0 is the capacitance of the sample measured in air.

$$\varepsilon' = \frac{C_{oil}}{C_{air}} = \frac{Im\left(Y_{oil}\right)}{Im\left(Y_{air}\right)} \tag{4}$$

$$\sigma = \frac{G\varepsilon_0}{C_0} = \frac{Im\left(Y_{oil}\right)\varepsilon_0}{Re\left(Y_{oil}\right)CC_0\omega} \tag{5}$$

Both chemical and electrical parameters increase with heating time. The results show a strong correlation between the changes of acidity and the conductivity (Fig. 4b) and between the changes of dielectric constant and the total polar components (Fig. 4c). The coefficient of correlation R^2 between the acidity and the conductivity is higher than 0.98, while it is higher than 0.69 between the total polar components and the dielectric constant for all type of oils. This confirms that both the real and imaginary parts of impedance are interesting for characterizing the oil as they deliver diverse information.

Conclusion

Impedance spectroscopy provides interesting possibilities for food quality assessment. This approach has been shown to be effective with regard to different food types, such as meat, milk and oil. In all cases, we have experimentally demonstrated that impedance measurements, which can be carried out in real time and are not restricted to laboratory environments, allow relevant information to be extracted about the state of the food, related to its conductivity behavior. The specially treated examples can be taken as representative for further food types or food mixtures, which have a primarily conductive, capacitive or structured material.

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